Visualization of saved distance matrices

Visualize arrays saved in `/res/sim_res` folder here.

TODO: Implement a SIS model

TODO: Do this in ELFI.

In [71]:
```
import grid_functions
importlib.reload(grid_functions)
from grid_functions import *

# Change theta_bsi quickly:
R_bsi = 3176/5400000/14 # proportion of BSI cases per year in the entire
theta_bsi = R_bsi/52 # proportion of BSI cases per week in the entire pop

modify_grid_params("grid/grid_params/final yearly bsi", "theta_bsi", thet
```

```
DirectoryError: grid/grid_params/final yearly bsi/.ipynb_checkpoints
DirectoryError: grid/grid_params/final yearly bsi/.ipynb_checkpoints
```

In [ ]:

In [16]:
```
import os
os.getcwd()

os.chdir('/u/50/ojalaf2/unix/Dropbox (Aalto)/Ecoli/')
```

In [17]:
```
print(os.getcwd())

from cluster.scripts.load_data import *
```

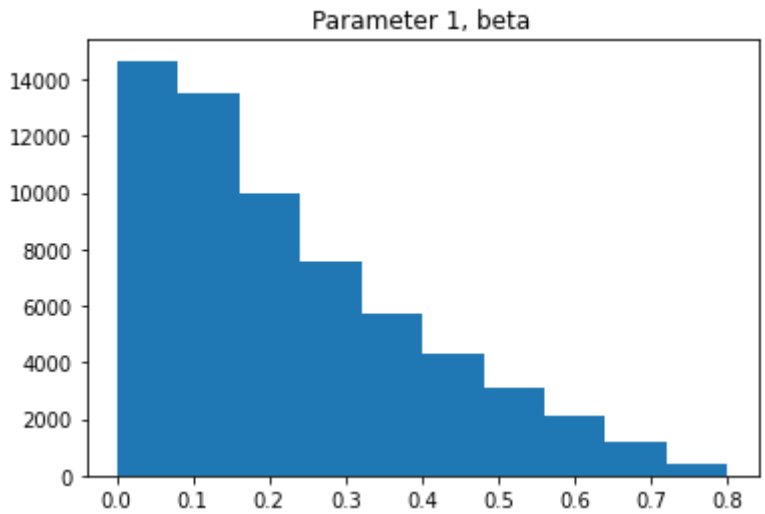
```
/u/50/ojalaf2/unix/Dropbox (Aalto)/Ecoli
```

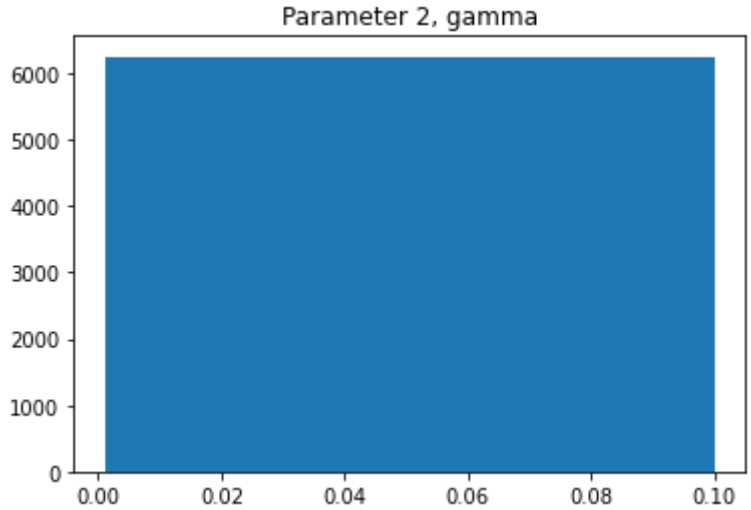
In [18]:
```
#!cp grid/grid_params/final/grid_params_A_I0.py grid/grid_params.py
```

In [19]:
```
#!python3 grid/run_grid.py
```

In [20]:
```
import importlib
import numpy as np
import matplotlib.pyplot as plt

from grid.grid_functions import *
from cluster.scripts.BSI_functions import *

incidence = True # If true, use incidence data instead of proportions.
```

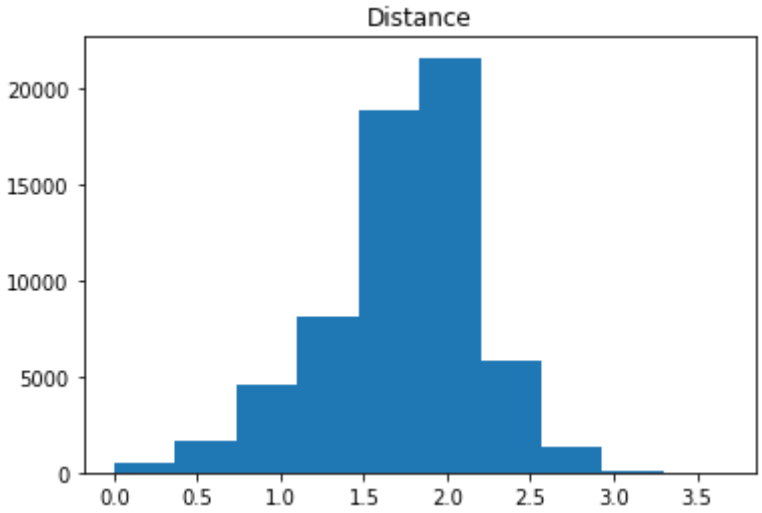
In [21]:
```
# How to visualize priors:
plt.hist(pairs[:,0])
plt.title("Parameter 1, beta") # or net transmission
plt.show()

plt.hist(pairs[:,1])
plt.title("Parameter 2, gamma") # or R
plt.show()
```



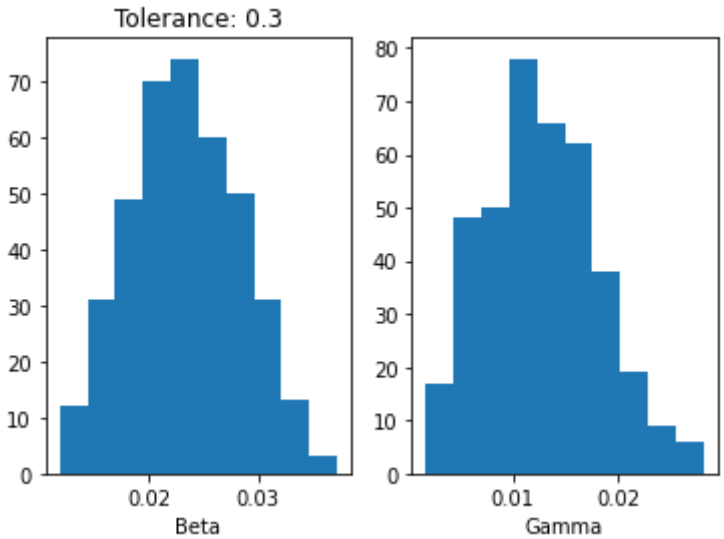


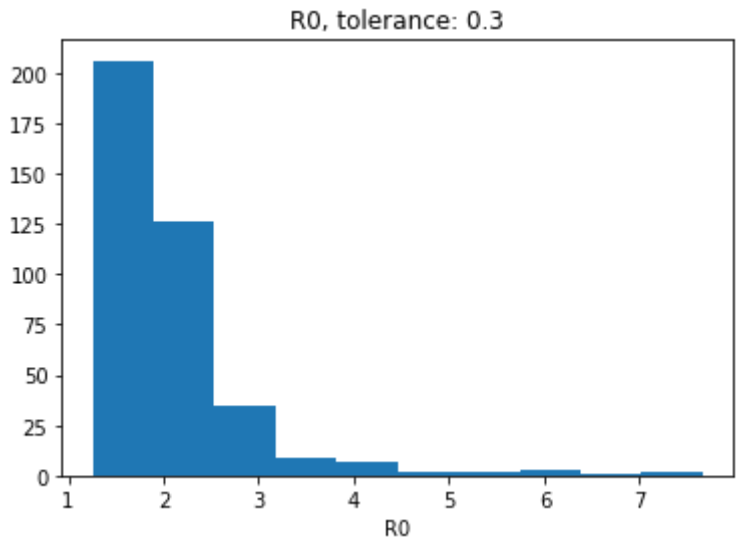
In [22]:
```
# NORM, clade A, no I0 fix

import grid_functions
importlib.reload(grid_functions)
from grid_functions import *

visualize_results("res/sim_res/time shift/obs_NORM_A_I0_2023-09-05_12-38-



visualize_results("res/sim_res/time shift/obs_NORM_A_no_I0_2023-09-05_12-
```

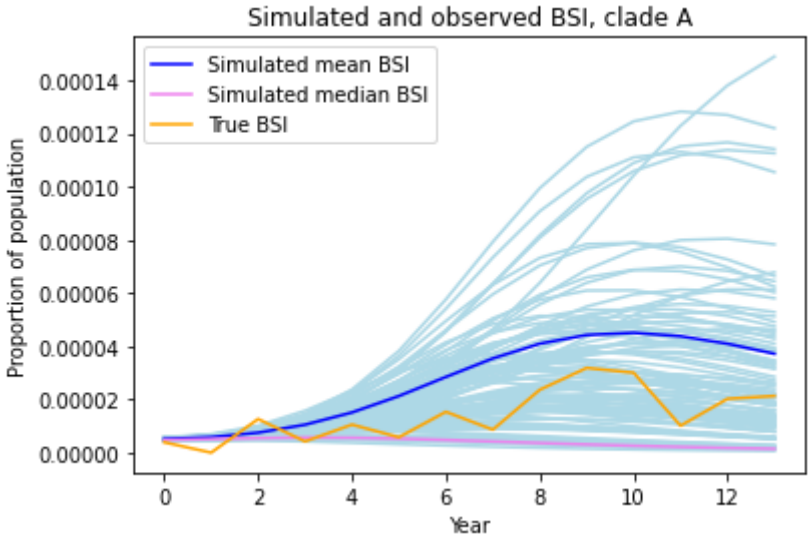
```
Results to visualize: res/sim_res/time shift/obs_NORM_A_I0_2023-09-05_12-
38-21/sim_params.txt
Year
2004    3.895481e-06
2005    3.949891e-10
2006    1.263574e-05
2007    4.214390e-06
2008    1.053623e-05
2009    5.791327e-06
2010    1.535184e-05
2011    8.640565e-06
2012    2.363202e-05
2013    3.185117e-05
2014    3.004447e-05
2015    1.012509e-05
2016    2.025119e-05
2017    2.130096e-05
Name: A, dtype: float64
```

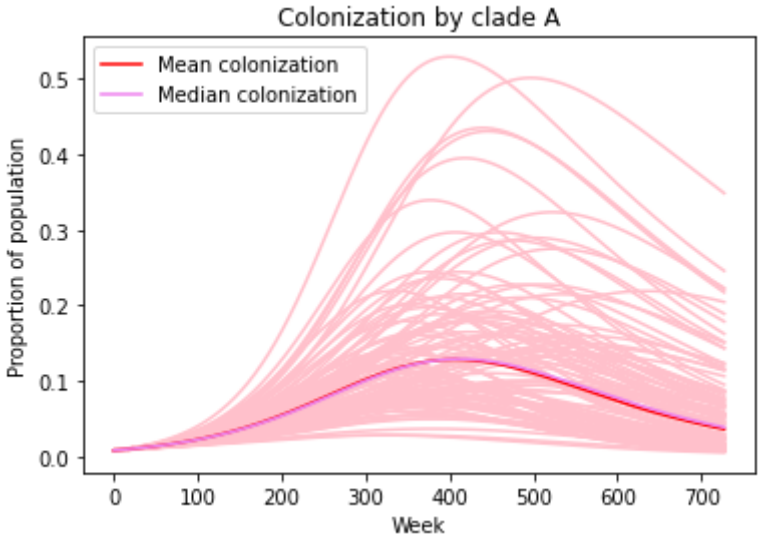


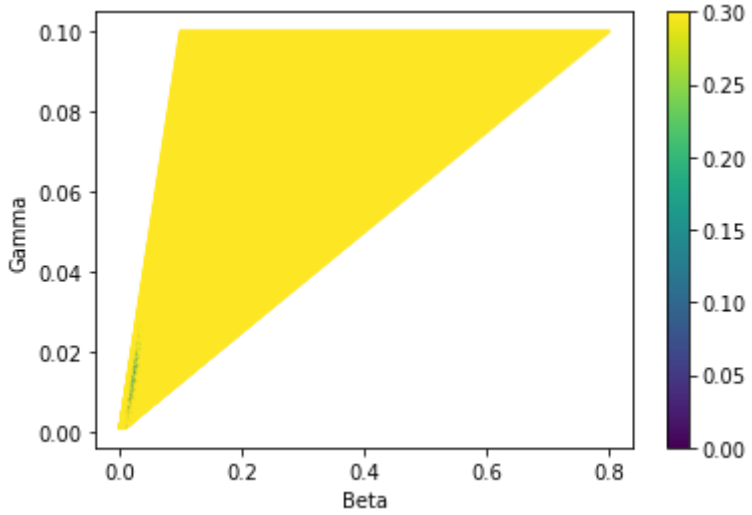

```
Beta mean: 0.023343700258948056
Gamma mean: 0.012748643428676536
R = mean(beta/gamma): 2.0933895962475857
R = median(beta/gamma): 1.8714946987951806
```



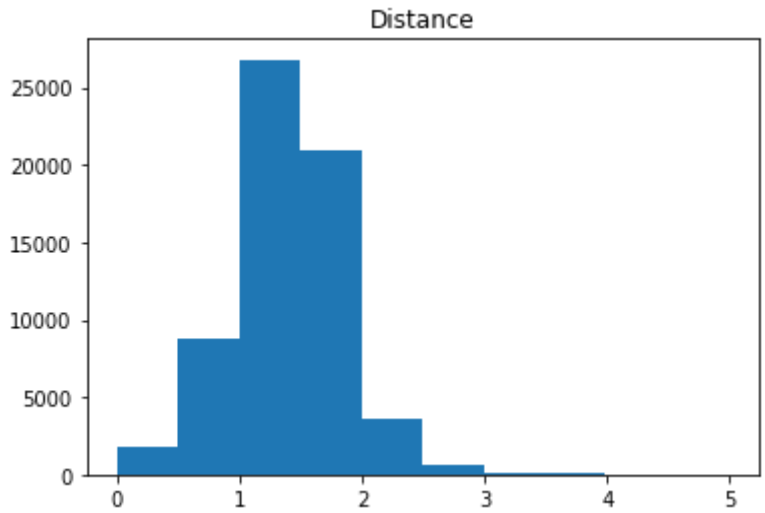

R0, tolerance: 0.3
200
175
150
125
100
75
50
25
0
1
2
3
4
5
6
7
R0


Simulated and observed BSI, clade A
Simulated mean BSI
Simulated median BSI
True BSI
0.00014
0.00012
0.00010
0.00008
0.00006
0.00004
0.00002
0.00000
Proportion of population
0
2
4
6
8
10
12
Year

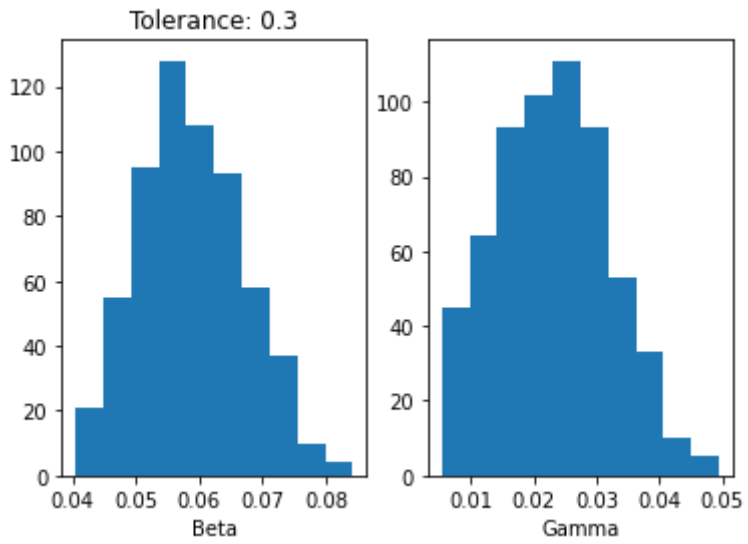

Colonization by clade A
Mean colonization
Median colonization
0.5
0.4
0.3
0.2
0.1
0.0
Proportion of population
0
100
200
300
400
500
600
700
Week

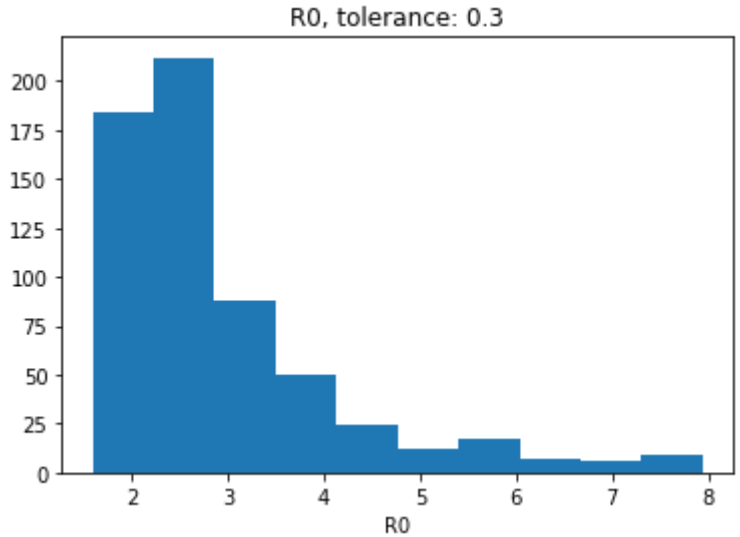


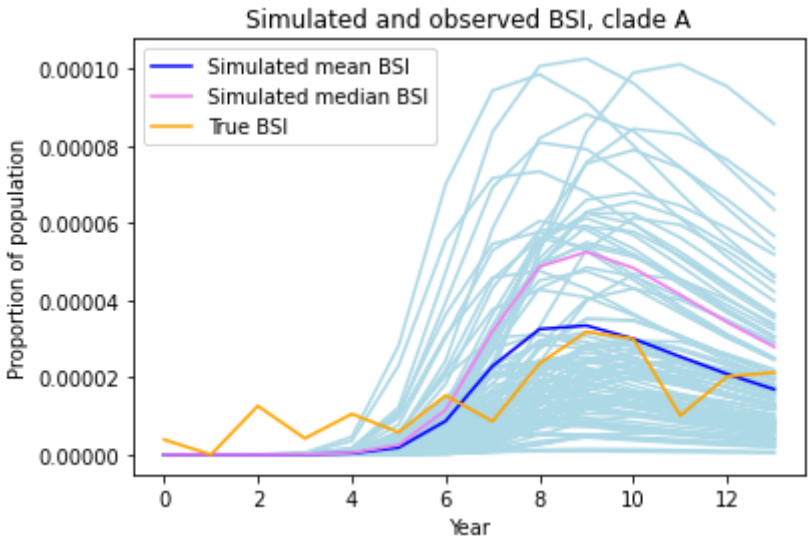

```
Results to visualize: res/sim_res/time shift/obs_NORM_A_no_I0_2023-09-05_
12-57-28/sim_params.txt
Year
2004    3.895481e-06
2005    3.949891e-10
2006    1.263574e-05
2007    4.214390e-06
2008    1.053623e-05
2009    5.791327e-06
2010    1.535184e-05
2011    8.640565e-06
2012    2.363202e-05
2013    3.185117e-05
2014    3.004447e-05
2015    1.012509e-05
2016    2.025119e-05
2017    2.130096e-05
Name: A, dtype: float64
```

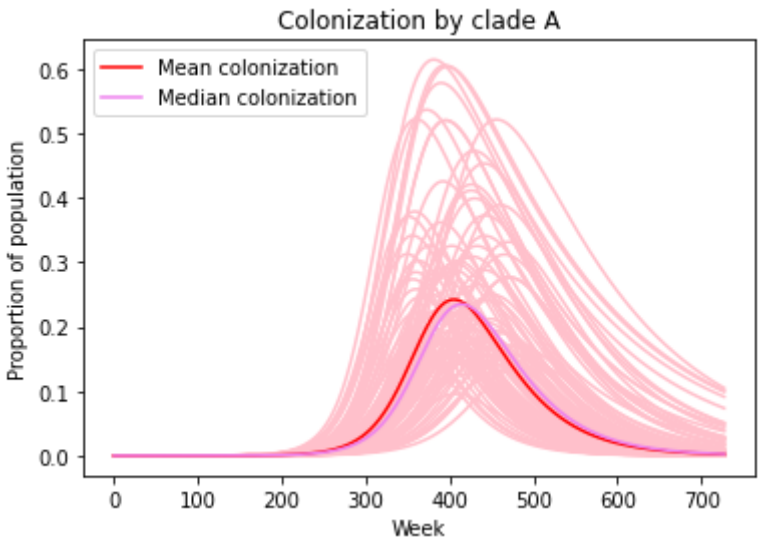


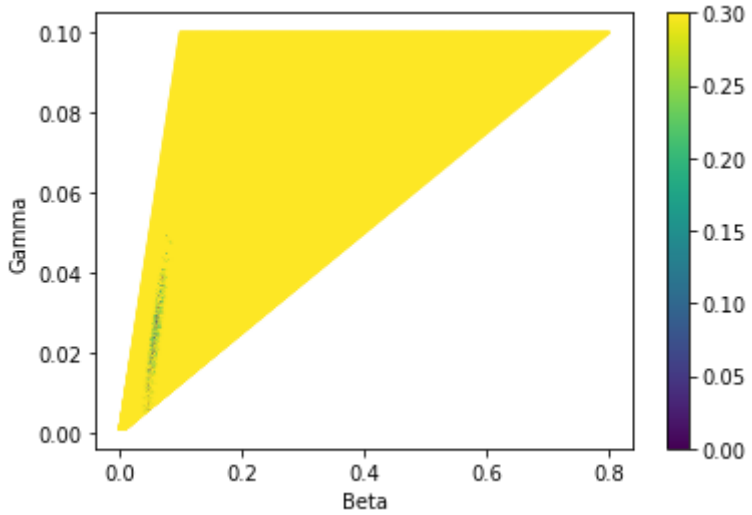

```
Beta mean: 0.058861344933402074
Gamma mean: 0.02311229153065464
R = mean(beta/gamma): 2.934226873998457
R = median(beta/gamma): 2.546192530120482
```


Tolerance: 0.3
Beta
Gamma


R0, tolerance: 0.3
R0

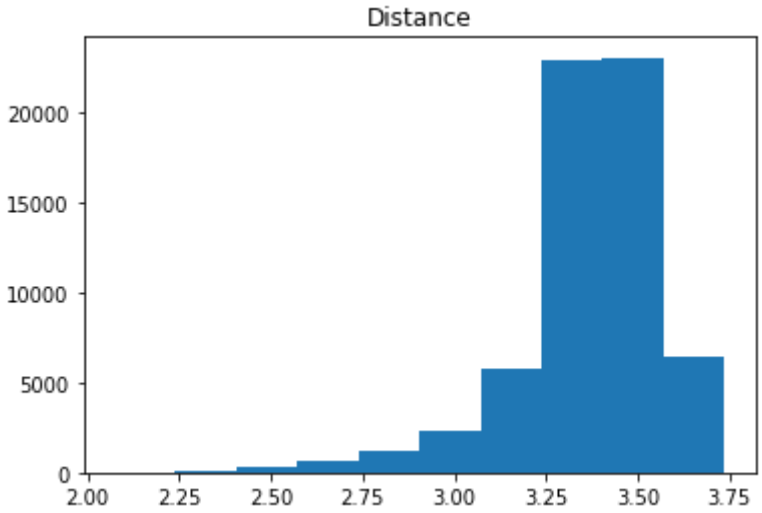

Simulated and observed BSI, clade A
Simulated mean BSI
Simulated median BSI
True BSI
Proportion of population
Year



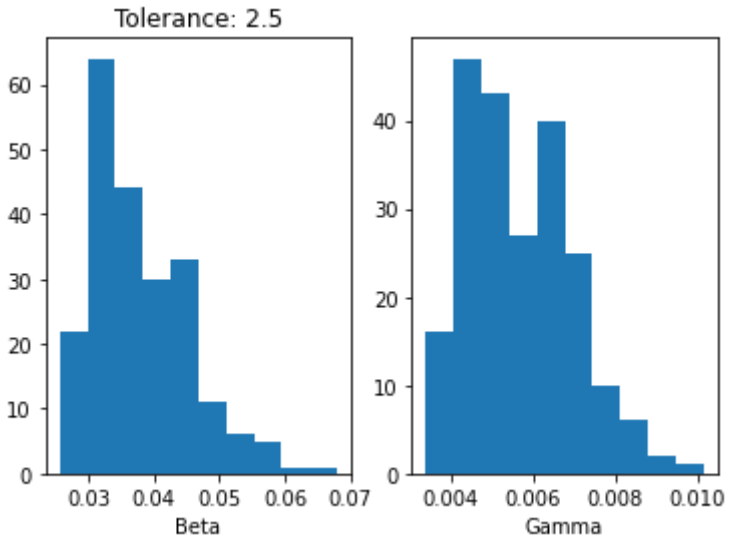


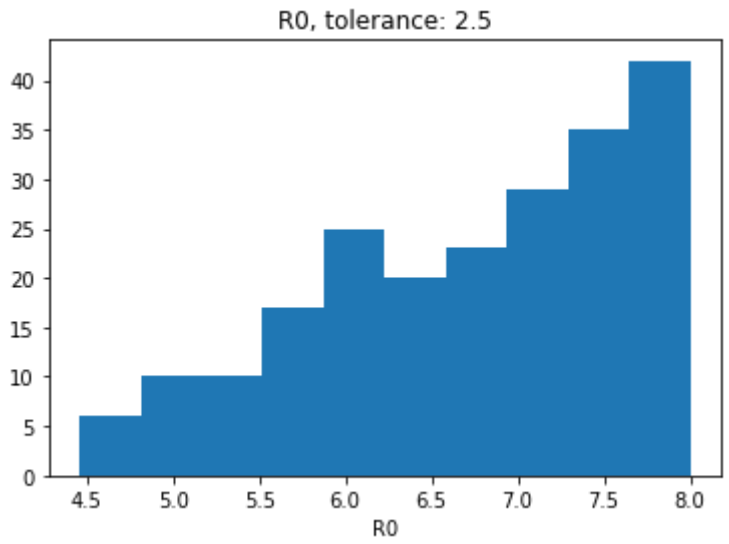
In [23]:
```
# NORM, clade C2, I0 fix

#visualize_results("res/sim_res/time shift/obs_NORM_C2_I0_2023-09-05_13-1



# NORM clade C2,no I0 fix (starts at 0)

visualize_results("res/sim_res/time shift/obs_NORM_C2_no_I0_2023-09-05_13

# Too restrictive prior?
```

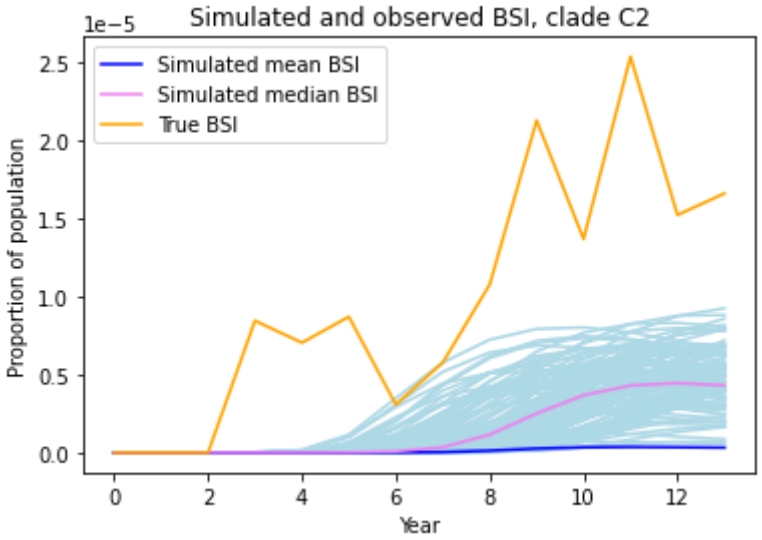
```
Results to visualize: res/sim_res/time shift/obs_NORM_C2_no_I0_2023-09-05
_13-35-57/sim_params.txt
Year
2004    3.895481e-10
2005    3.949891e-10
2006    4.211914e-10
2007    8.428780e-06
2008    7.024155e-06
2009    8.686990e-06
2010    3.070367e-06
2011    5.760377e-06
2012    1.074183e-05
2013    2.123411e-05
2014    1.365658e-05
2015    2.531273e-05
2016    1.518839e-05
2017    1.656741e-05
Name: C2, dtype: float64
```

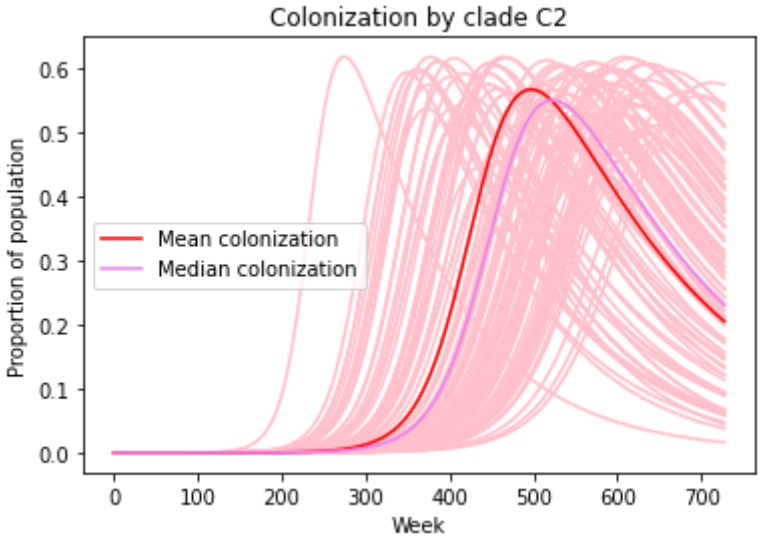


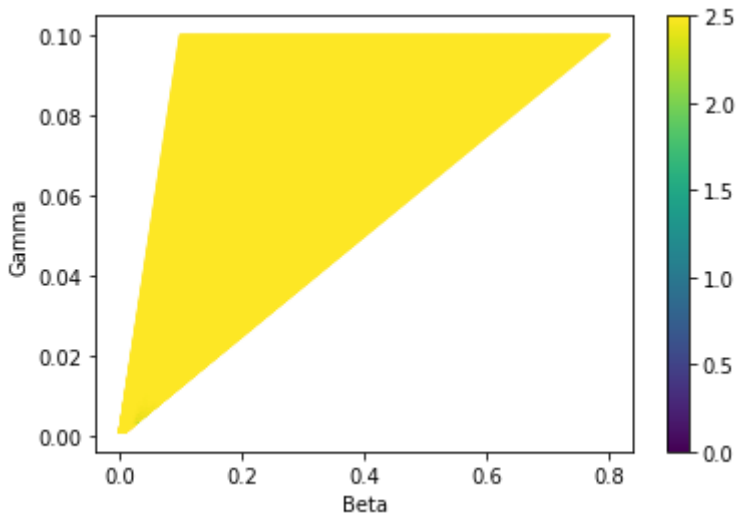

```
Beta mean: 0.03774079534386951
Gamma mean: 0.005688634723224696
R = mean(beta/gamma): 6.737273471212038
R = median(beta/gamma): 6.875503614457833
```




R0, tolerance: 2.5
0
5
10
15
20
25
30
35
40
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
R0

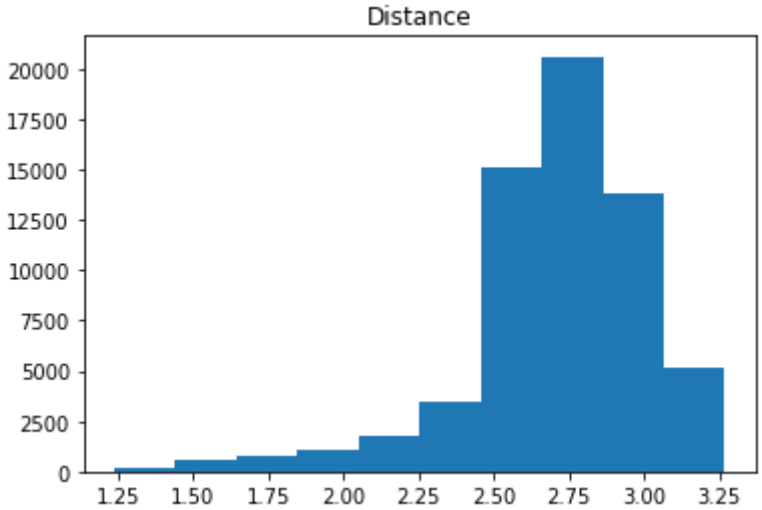

Simulated and observed BSI, clade C2
1e−5
Simulated mean BSI
Simulated median BSI
True BSI
Proportion of population
0.0
0.5
1.0
1.5
2.0
2.5
0
2
4
6
8
10
12
Year


Colonization by clade C2
Mean colonization
Median colonization
Proportion of population
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
100
200
300
400
500
600
700
Week

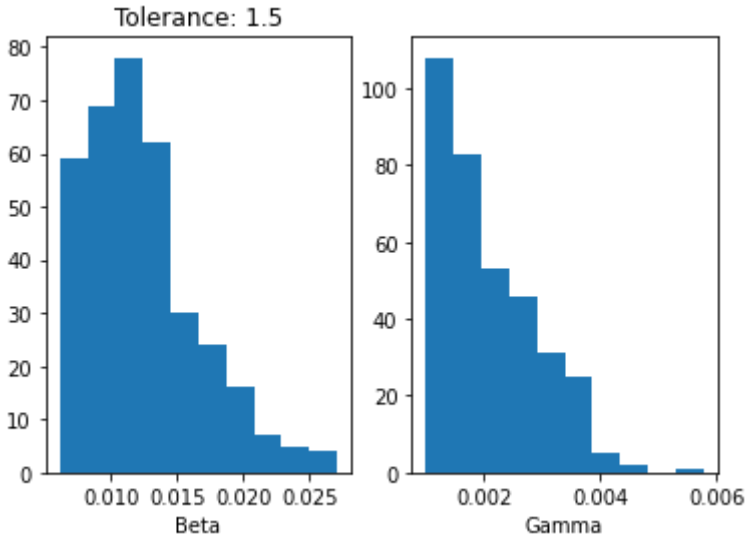


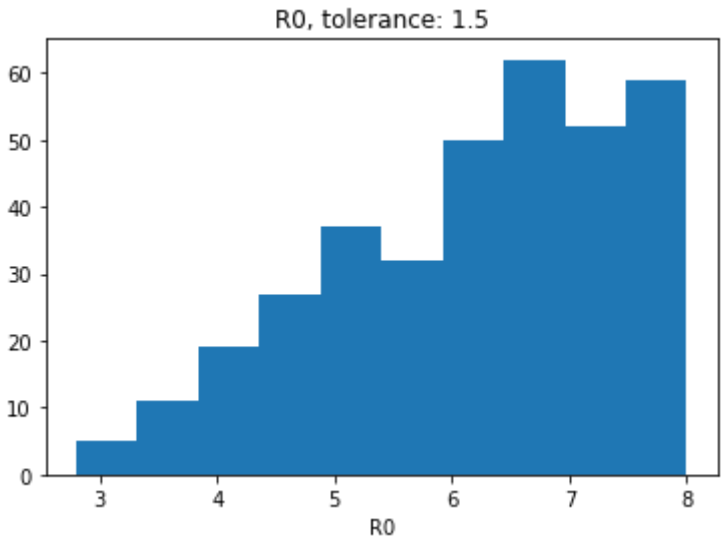
In [24]:
```
import grid_functions
importlib.reload(grid_functions)
from grid_functions import *


# NORM, clade C1:
visualize_results("res/sim_res/time shift/obs_NORM_C1_I0_2023-09-05_13-17

# NORM, clade C1, no I0:
visualize_results("res/sim_res/time shift/obs_NORM_C1_no_I0_2023-09-05_13
```

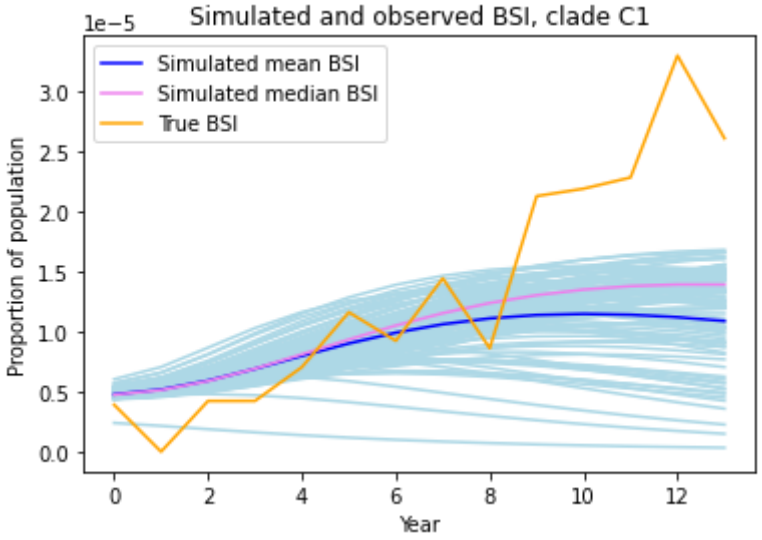
```
Results to visualize: res/sim_res/time shift/obs_NORM_C1_I0_2023-09-05_13
-17-42/sim_params.txt
Year
2004    3.895481e-06
2005    3.949891e-10
2006    4.211914e-06
2007    4.214390e-06
2008    7.024155e-06
2009    1.158265e-05
2010    9.211101e-06
2011    1.440094e-05
2012    8.593462e-06
2013    2.123411e-05
2014    2.185052e-05
2015    2.278146e-05
2016    3.290818e-05
2017    2.603450e-05
Name: C1, dtype: float64
```

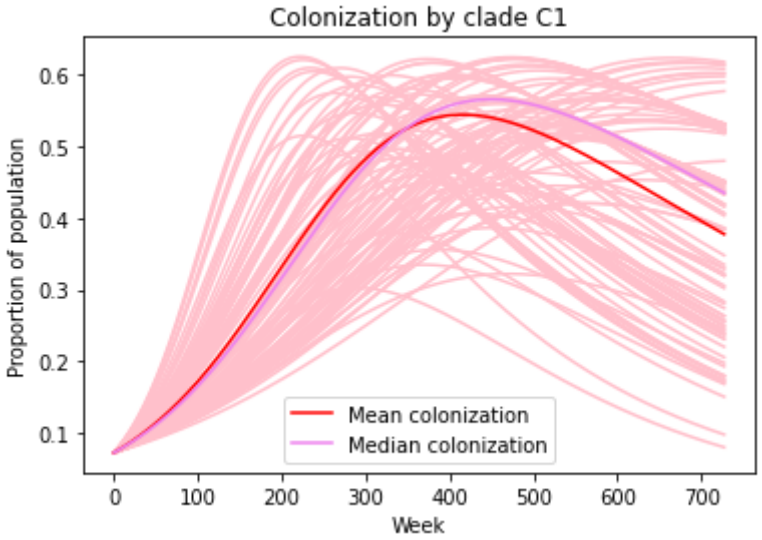


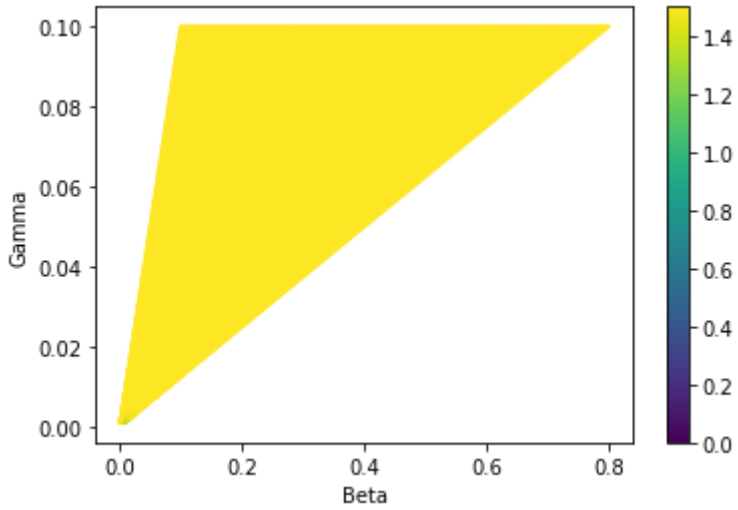

```
Beta mean: 0.01232431842438162
Gamma mean: 0.002077088013069226
R = mean(beta/gamma): 6.1654667371179634
R = median(beta/gamma): 6.397592650602411
```



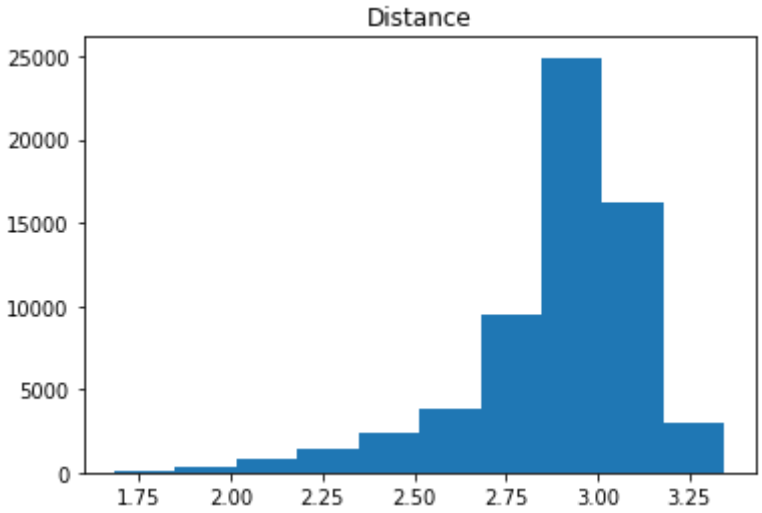



Simulated and observed BSI, clade C1
1e−5
Simulated mean BSI
Simulated median BSI
True BSI
Proportion of population
Year

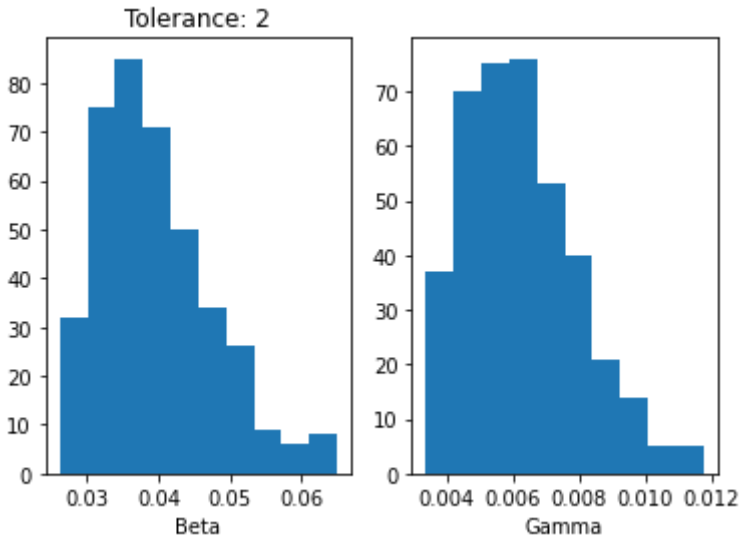

Colonization by clade C1
Mean colonization
Median colonization
Proportion of population
Week

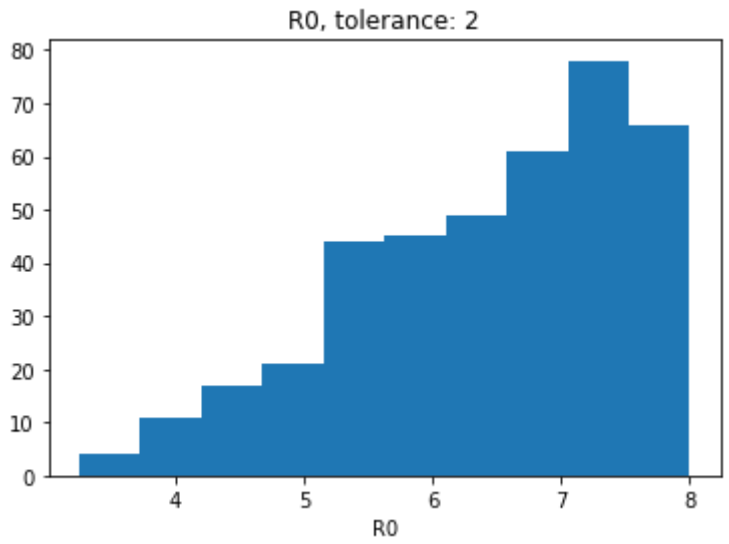

Gamma
Beta

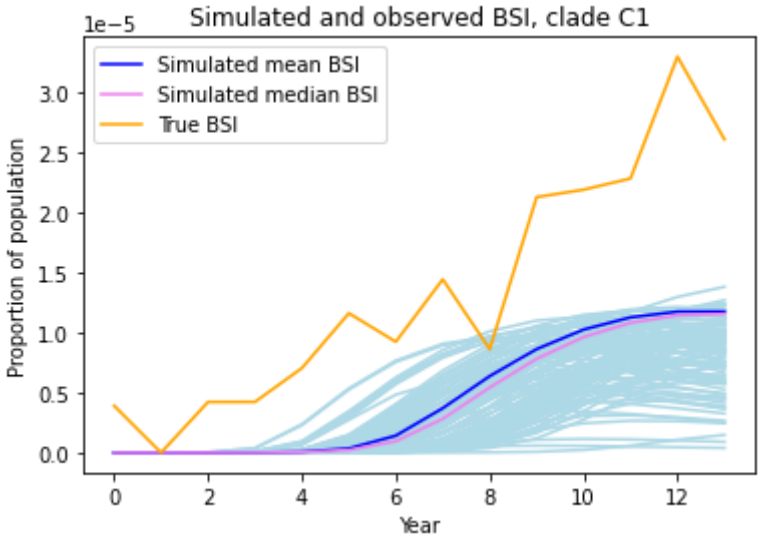
```
Results to visualize: res/sim_res/time shift/obs_NORM_C1_no_I0_2023-09-05
_13-38-05/sim_params.txt
Year
2004    3.895481e-06
2005    3.949891e-10
2006    4.211914e-06
2007    4.214390e-06
2008    7.024155e-06
2009    1.158265e-05
2010    9.211101e-06
2011    1.440094e-05
2012    8.593462e-06
2013    2.123411e-05
2014    2.185052e-05
2015    2.278146e-05
2016    3.290818e-05
2017    2.603450e-05
Name: C1, dtype: float64
```

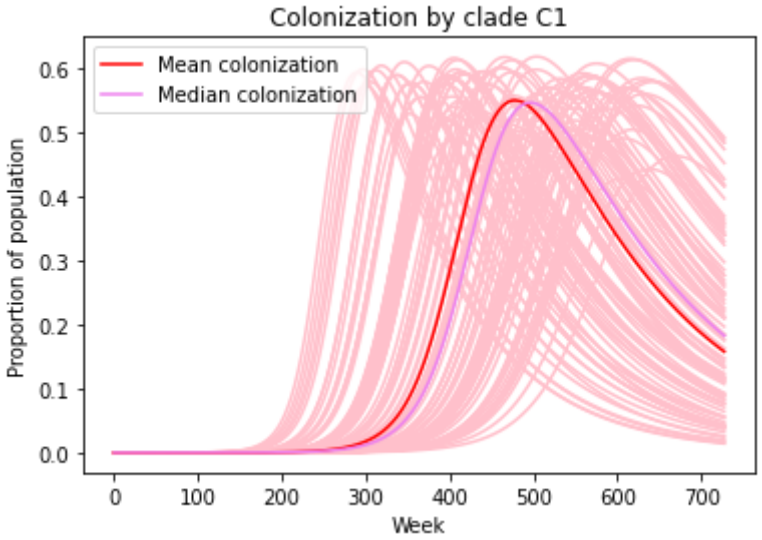


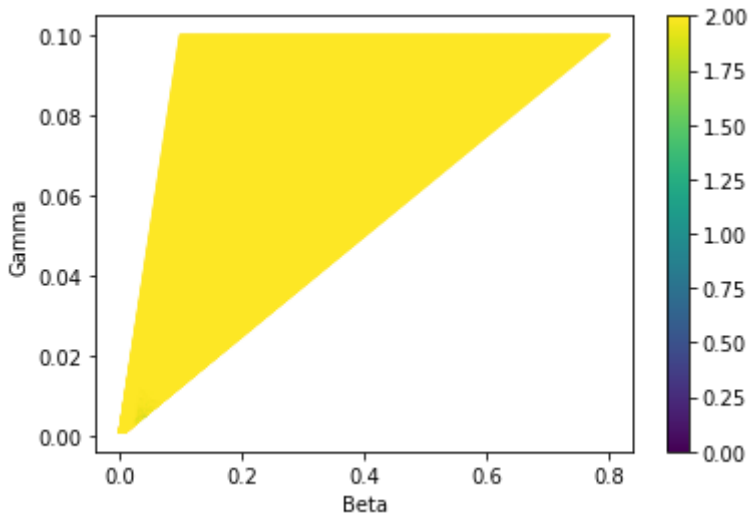

```
Beta mean: 0.039522004809588854
Gamma mean: 0.006322289156626506
R = mean(beta/gamma): 6.429964516246805
R = median(beta/gamma): 6.6365481325301205
```




R0, tolerance: 2
80
70
60
50
40
30
20
10
0
4
5
6
7
8
R0

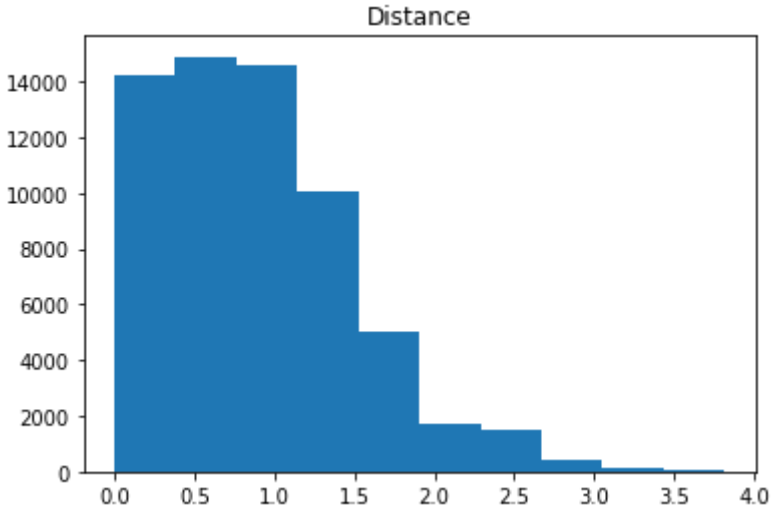

Simulated and observed BSI, clade C1
1e−5
Simulated mean BSI
Simulated median BSI
True BSI
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Proportion of population
0
2
4
6
8
10
12
Year


Colonization by clade C1
Mean colonization
Median colonization
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Proportion of population
0
100
200
300
400
500
600
700
Week

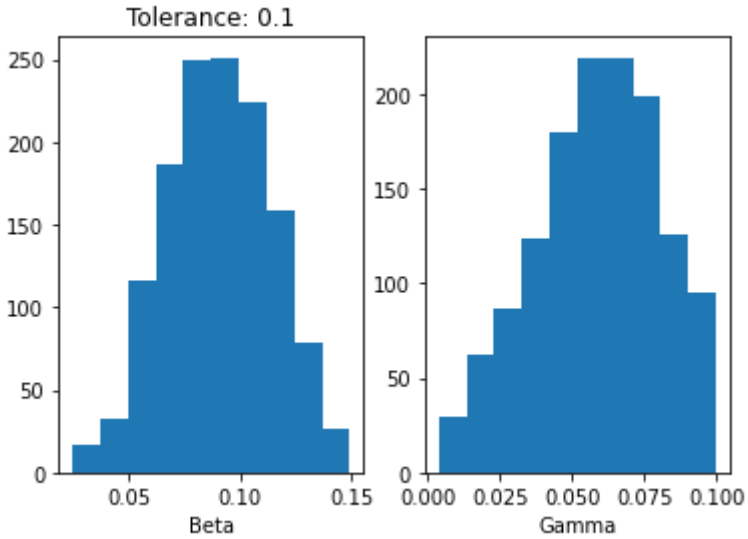


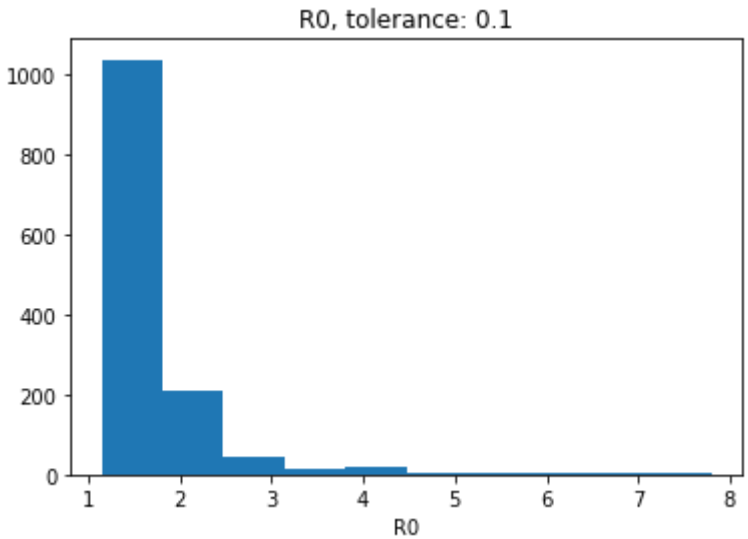
In [25]:
```
import grid_functions
importlib.reload(grid_functions)
from grid_functions import *


# NORM, clade B:
visualize_results("res/sim_res/time shift/obs_NORM_B_I0_2023-09-05_12-38-

# NORM, clade B, no I0:
visualize_results("res/sim_res/time shift/obs_NORM_B_no_I0_2023-09-05_12-
```

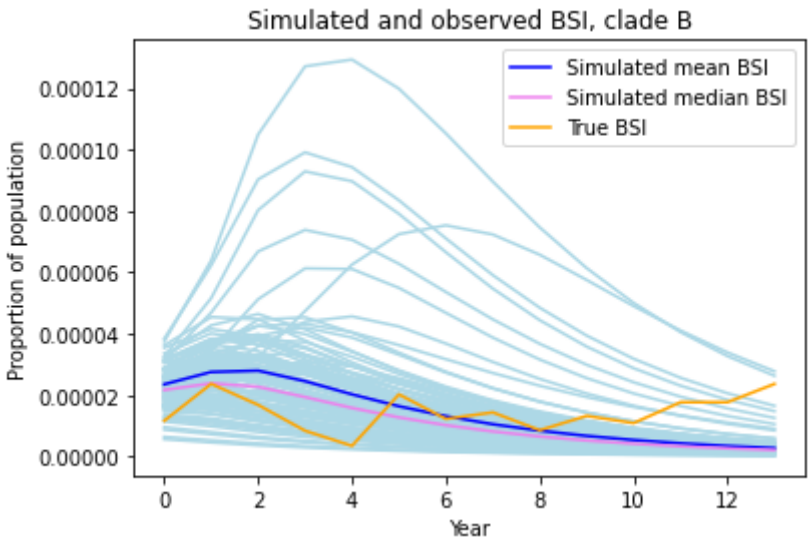
```
Results to visualize: res/sim_res/time shift/obs_NORM_B_I0_2023-09-05_12-
38-26/sim_params.txt
Year
2004    0.000012
2005    0.000024
2006    0.000017
2007    0.000008
2008    0.000004
2009    0.000020
2010    0.000012
2011    0.000014
2012    0.000009
2013    0.000013
2014    0.000011
2015    0.000018
2016    0.000018
2017    0.000024
Name: B, dtype: float64
```

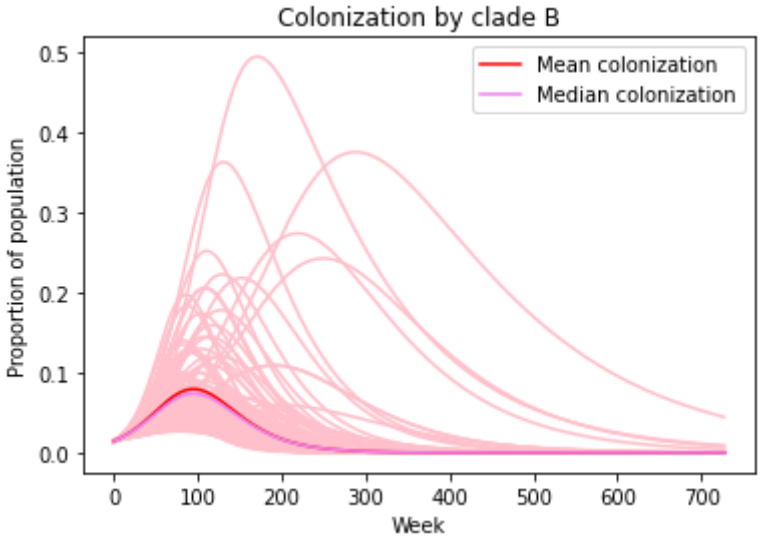


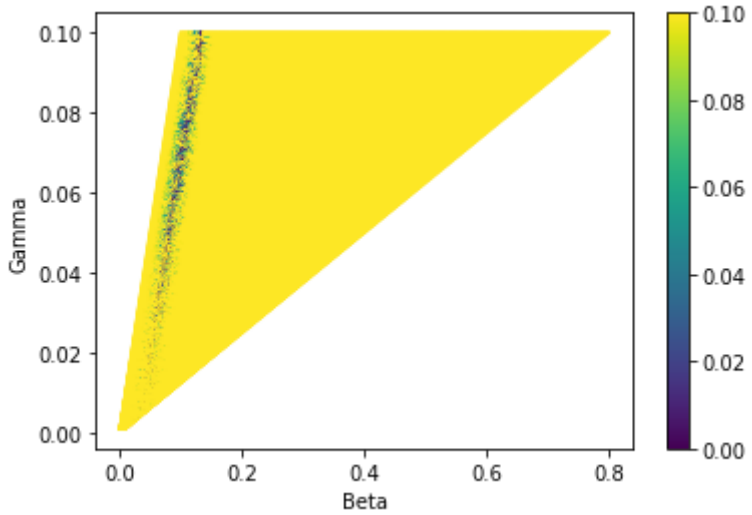

```
Beta mean: 0.09034715085165299
Gamma mean: 0.05888351915123179
R = mean(beta/gamma): 1.6954774474015466
R = median(beta/gamma): 1.5060333734939764
```



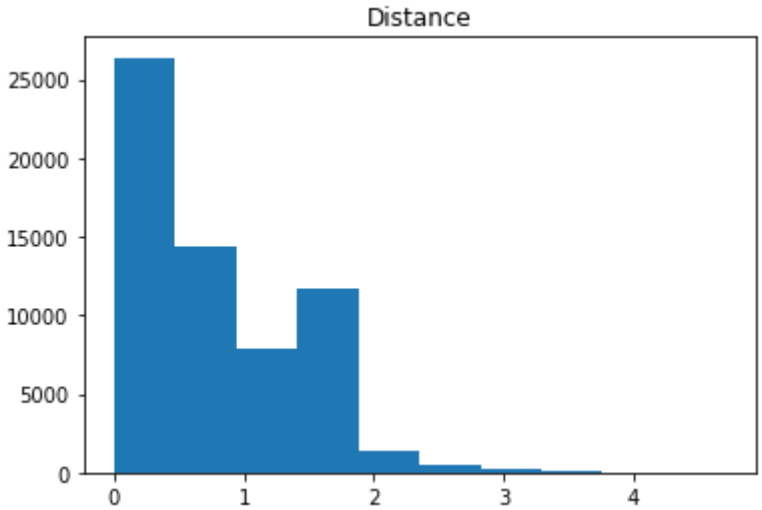



Simulated and observed BSI, clade B
Simulated mean BSI
Simulated median BSI
True BSI
Proportion of population
0.00012
0.00010
0.00008
0.00006
0.00004
0.00002
0.00000
0
2
4
6
8
10
12
Year

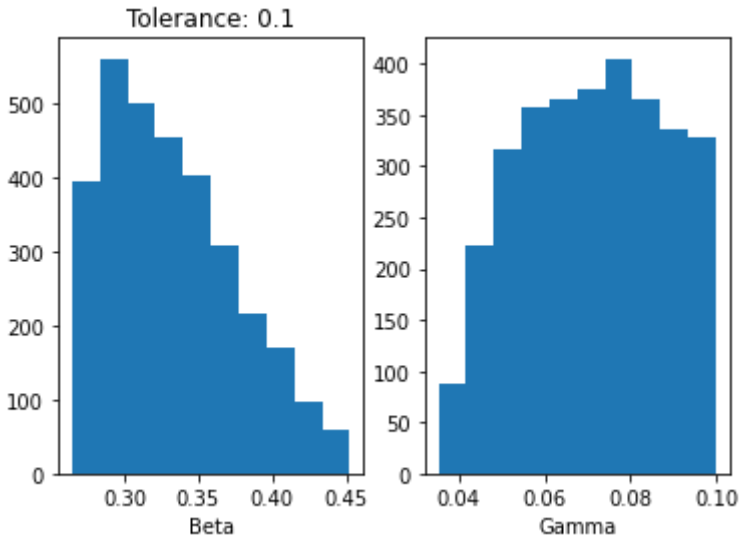

Colonization by clade B
Mean colonization
Median colonization
Proportion of population
0.5
0.4
0.3
0.2
0.1
0.0
0
100
200
300
400
500
600
700
Week

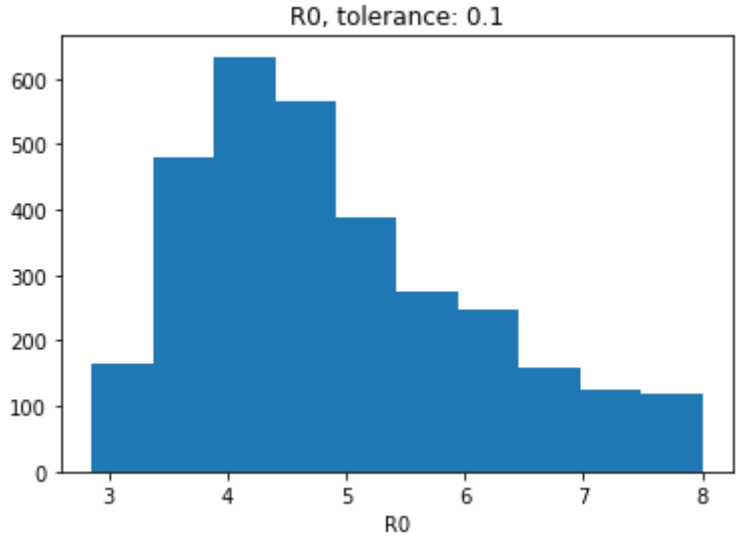

Gamma
0.10
0.08
0.06
0.04
0.02
0.00
0.0
0.2
0.4
0.6
0.8
Beta
0.10
0.08
0.06
0.04
0.02
0.00

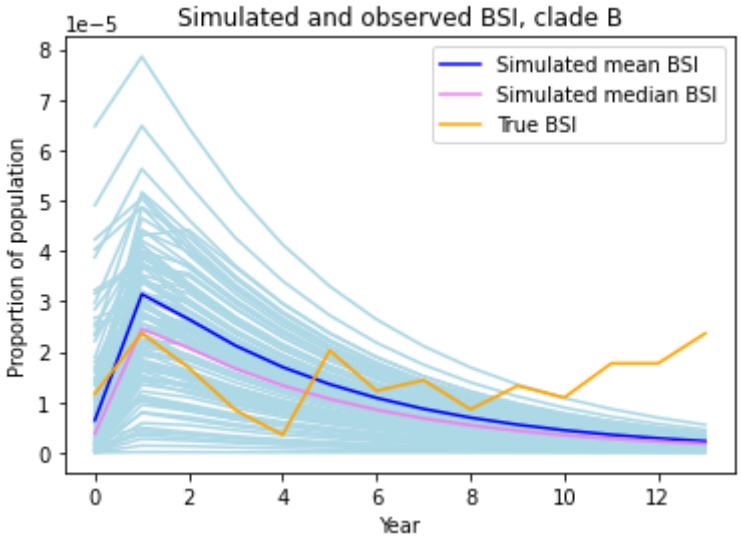
```
Results to visualize: res/sim_res/time shift/obs_NORM_B_no_I0_2023-09-05_
12-58-17/sim_params.txt
Year
2004    0.000012
2005    0.000024
2006    0.000017
2007    0.000008
2008    0.000004
2009    0.000020
2010    0.000012
2011    0.000014
2012    0.000009
2013    0.000013
2014    0.000011
2015    0.000018
2016    0.000018
2017    0.000024
Name: B, dtype: float64
```

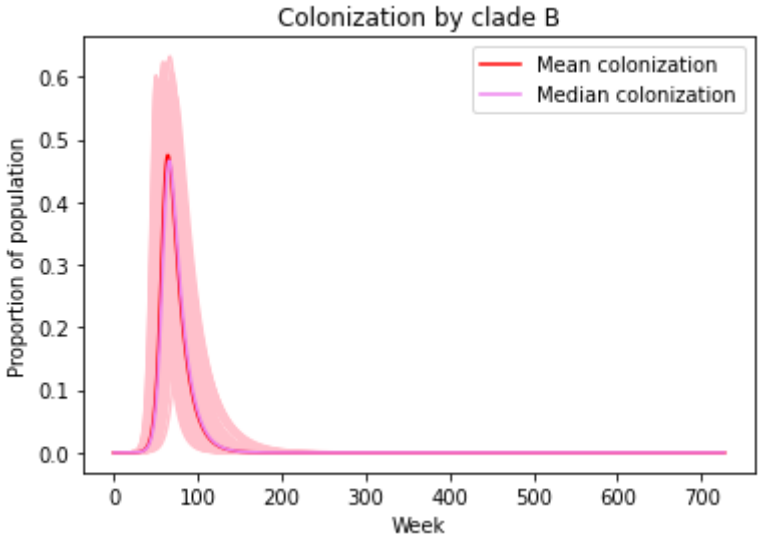


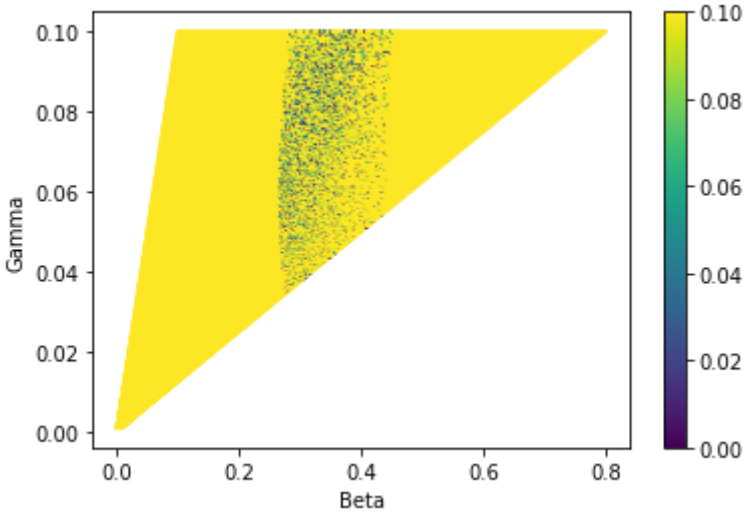

```
Beta mean: 0.33261905048045454
Gamma mean: 0.07107775313778836
R = mean(beta/gamma): 4.91228477854492
R = median(beta/gamma): 4.654623253012049
```




R0, tolerance: 0.1
600
500
400
300
200
100
0
3
4
5
6
7
8
R0


Simulated and observed BSI, clade B
1e−5
Simulated mean BSI
Simulated median BSI
True BSI
Proportion of population
Year


Colonization by clade B
Mean colonization
Median colonization
Proportion of population
Week


0.10
0.08
0.06
0.04
0.02
0.00
Gamma
0.0
0.2
0.4
0.6
0.8
Beta
0.10
0.08
0.06
0.04
0.02
0.00